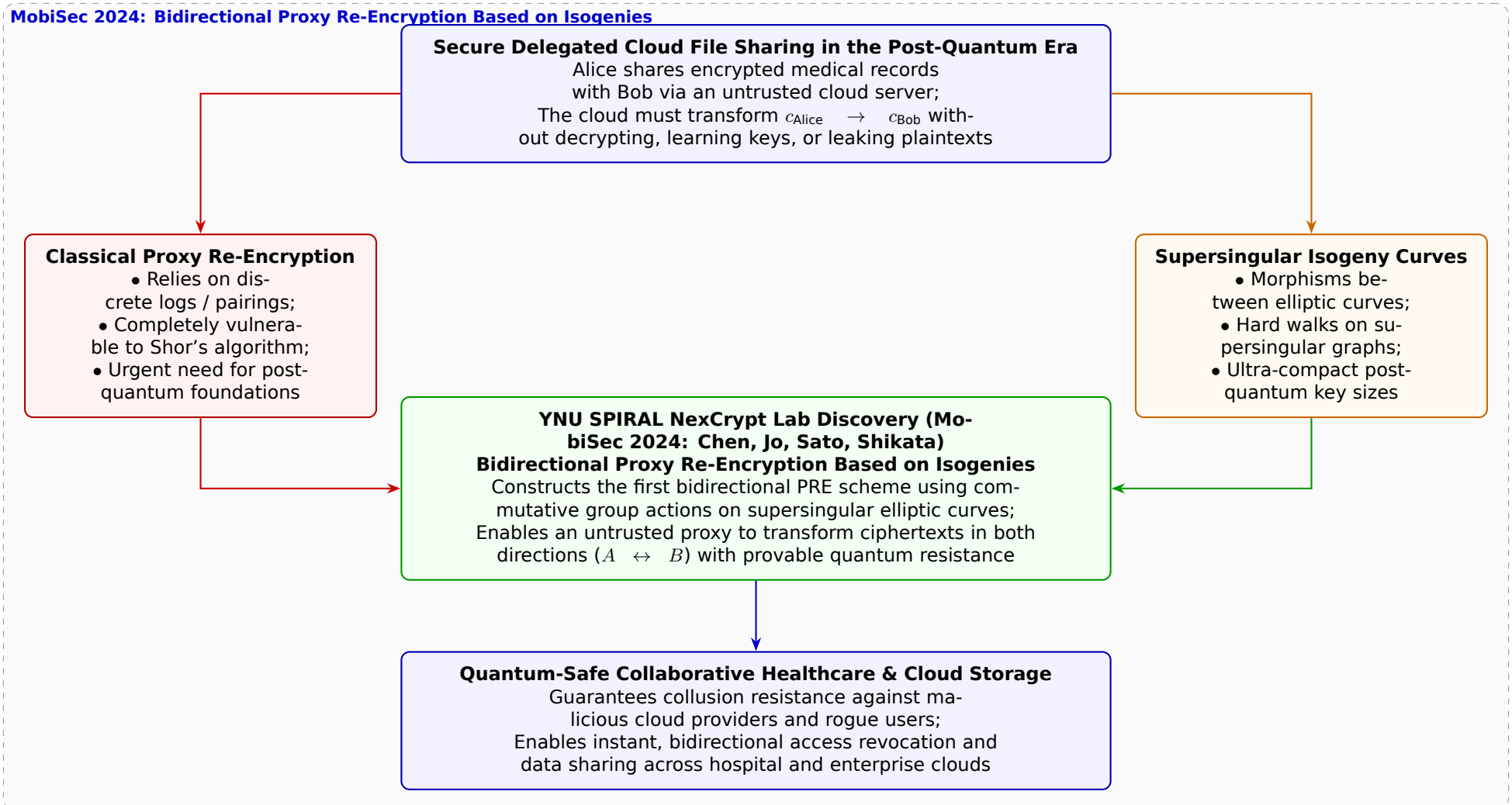
MobiSec 2024: Bidirectional Proxy Re-Encryption Based on Isogenies
Secure Delegated Cloud File Sharing in the Post-Quantum Era
Alice shares encrypted medical records with Bob via an untrusted cloud server; The cloud must transform $c_{\text{Alice}} \rightarrow c_{\text{Bob}}$ without decrypting, learning keys, or leaking plaintexts
Classical Proxy Re-Encryption
• Relies on discrete logs / pairings;
• Completely vulnerable to Shor's algorithm;
• Urgent need for post-quantum foundations
Supersingular Isogeny Curves
• Morphisms between elliptic curves;
• Hard walks on supersingular graphs;
• Ultra-compact post-quantum key sizes
YNU SPIRAL NexCrypt Lab Discovery (MobiSec 2024: Chen, Jo, Sato, Shikata)
Bidirectional Proxy Re-Encryption Based on Isogenies
Constructs the first bidirectional PRE scheme using commutative group actions on supersingular elliptic curves; Enables an untrusted proxy to transform ciphertexts in both directions ($A \leftrightarrow B$) with provable quantum resistance
Quantum-Safe Collaborative Healthcare & Cloud Storage
Guarantees collusion resistance against malicious cloud providers and rogue users; Enables instant, bidirectional access revocation and data sharing across hospital and enterprise clouds